



Carbon Pricing & Electricity Demand:

Did electricity demand respond more strongly to the Australian carbon pricing policy in states with more carbon-intensive electricity generation?

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WHAT Happened



Australia Introduced Carbon Pricing **1 July 2012 – 7 July 2014**

Australia introduces controversial carbon tax

© 1 July 2012



The law forces about 300 of the worst-polluting firms to pay a A\$23/ton of GHG emissions.

Households expected to pay ~10% more for electricity

Australia kills off carbon tax

Senate makes good on prime minister's 'pledge in blood' to 'axe the tax', with nation unlikely to meet 5% emissions reduction target

Lenore Taylor, *political editor*

Thu 17 Jul 2014 11.51 AEST

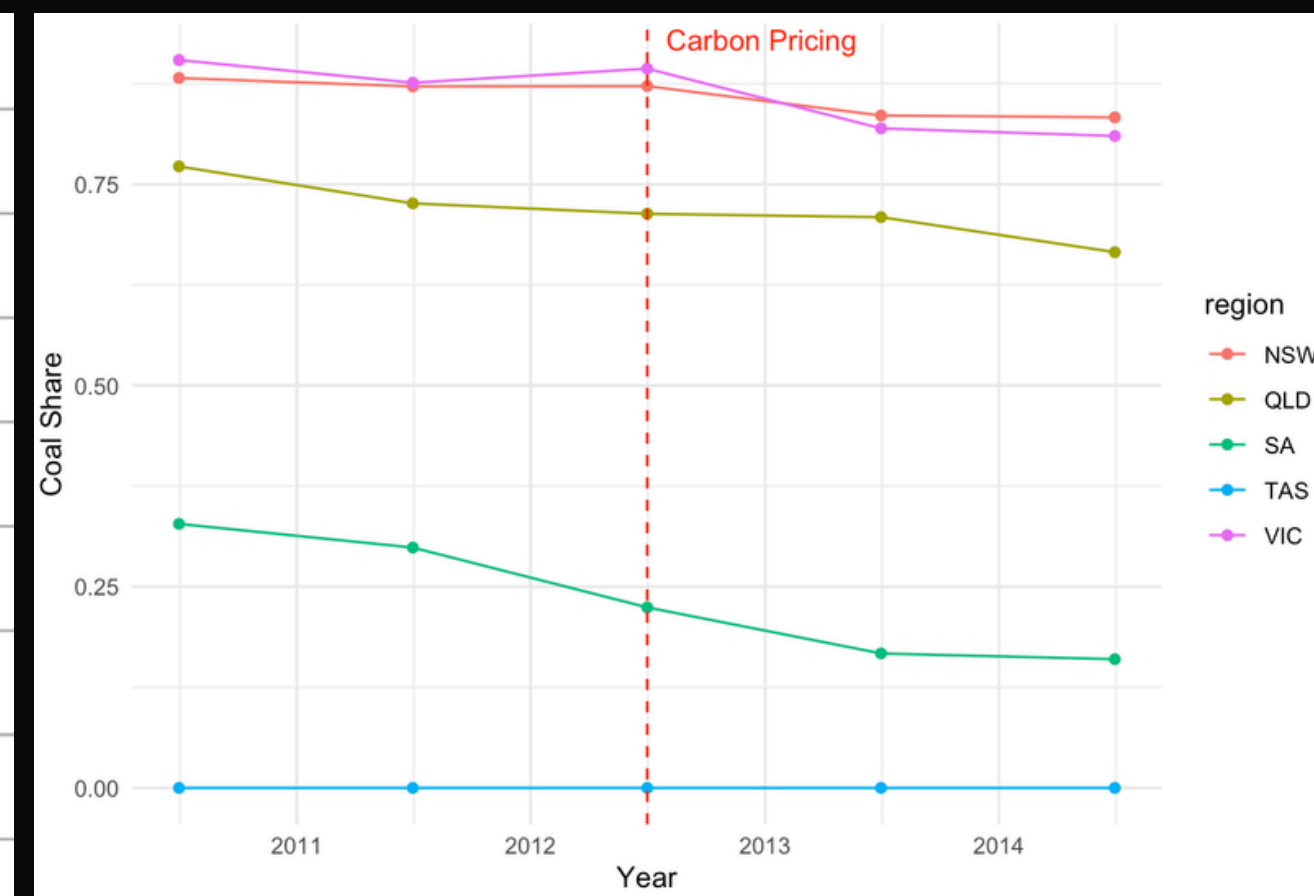
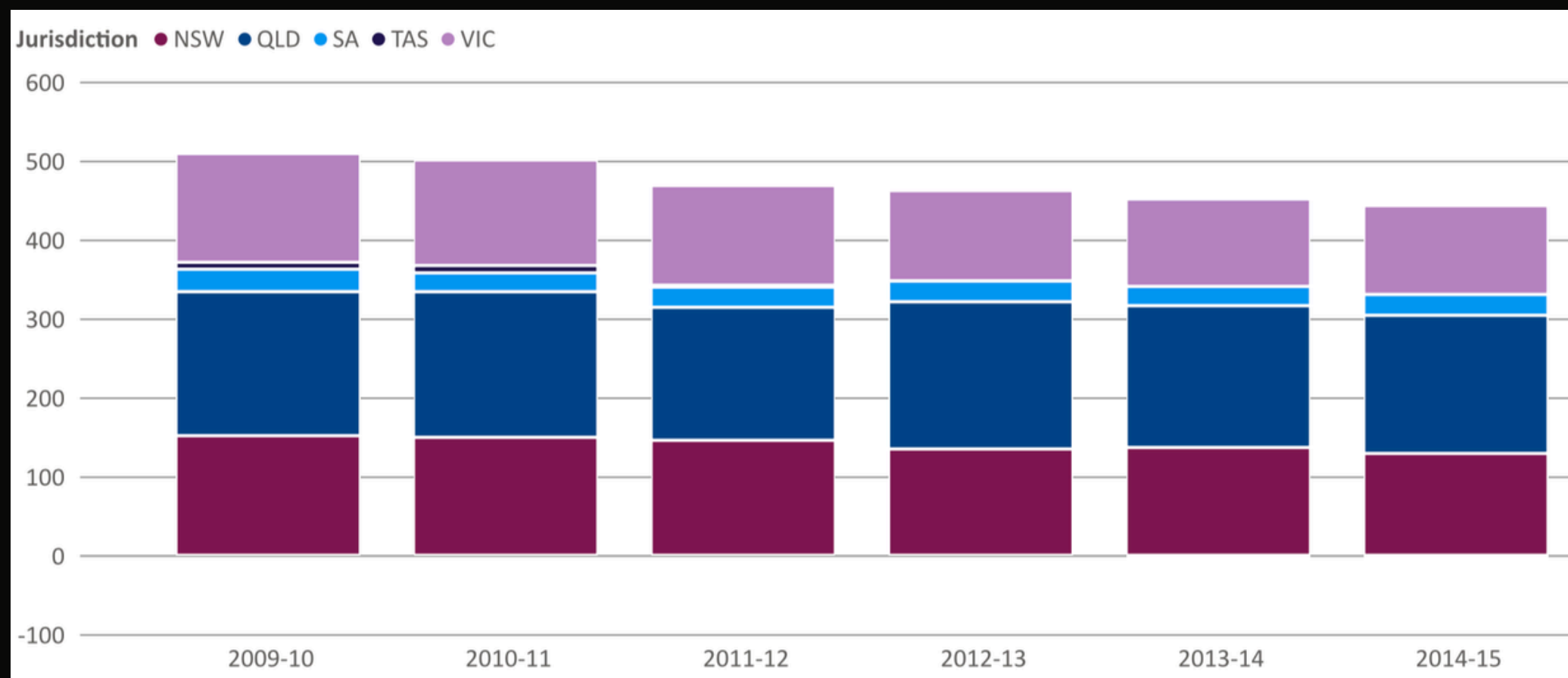
Electricity price did not revert to pre-CPM levels (Nazifi et al., 2011)

Carbon emissions resumed growth after repeal

Emissions trading scheme was introduced in July 2015



Each State's Grid Mix Shaped Its Exposure to Carbon Pricing



Total NEM emissions declined after 2012, with the largest absolute reductions in NSW and VIC — Australia's most coal-intensive states.

Coal share varies dramatically across states and declines slowly — TAS at zero throughout, NSW and VIC consistently above 0.75.

WHAT
The Literature Found

Evidences shows Carbon Pricing affecting **Electricity Price, Market Behavior, and Energy Supply Mix**

Price Pass-Through

Nazifi et al. (2021)

Carbon cost pass-through to spot electricity prices was 100 (VIC) –290% (SA) across NEM regions. Consumers, not generators, bore the costs. Minor reduction in emission intensities found.

Price and Market Behaviour

Wong & Zhang (2022)

Wholesale electricity costs rose 22–68% across connected regions after CPM. Significant difference in price impacts attributed to distinct fuel mix and access to renewable hydro.

Generation-side Adjustment

O’Gorman & Jotzo (2014)

Demand reduction rose from 2.5 to 4.2 TWh per year. Shifts in the supply mix resulted in a 16 to 28kg CO₂/MWh reduction in the emissions intensity of power supply.

Existing work focuses primarily on cost pass-through, electricity price, and generation-side adjustments rather than on heterogeneous demand responses across states with different carbon intensities.



WHAT is Missing?

Did electricity demand respond more strongly to the Australian carbon pricing policy in states with more carbon-intensive electricity generation?



The Data & Empirical Strategy



The sample was restricted to **NEM states within 2010 – 2014 time window**

- Australian Energy Market Operation (AEMO)
- Department of Climate Change, Energy, the Environment, and Water (DCCEEW)
- Australian Bureau of Meteorology (BOM)
- Australian Bureau of Statistics (ABS)

	Observation	Detail
RAw 30-min AEMO dispatch data, 2010–2014	438,240	
Aggregate to monthly state average	305	Aggregated, Drop None
Merge with energy mix, 2010 – 2014	300	Drop 5 (12 a.m. 2015 data)
Merge with temperature data, 2010 –2014	300	Drop None
Merge with quarter population data	100	Drop 200 (non-quarter months)
Analysis sample	100	

States outside National Electricity Market (NEM) are excluded because of different market mechanisms. Restrict the sample within NEM to ensure comparability and balance

Time window 2010–2014 creates symmetrical pre- and post-policy bounds. Avoiding mix weight on different regimes policies.



Descriptive analysis : no untreated control group exist

$$\log(demand_{it}) = \alpha_i + \lambda_t + \beta_1(post_t \times coalshare_i) + \beta_2 post_t + \gamma_1 temp_{it} + \gamma_2 temp_{it}^2 + \gamma_3 \log(pop_{it}) + \varepsilon_{it}$$

i indexes state, t indexes quarter-year.

α_i are **state fixed effects**, absorbing all time-invariant state-level differences

λ_t are **time fixed effects**, absorbing common shocks in each period

$post_t$ is a binary indicator equal to 1 from July 2012 onward

$post_t \times coalshare_i$ is the key variable of interest. It captures whether the post-policy change in demand was systematically larger in states with a higher share of coal in electricity generation.

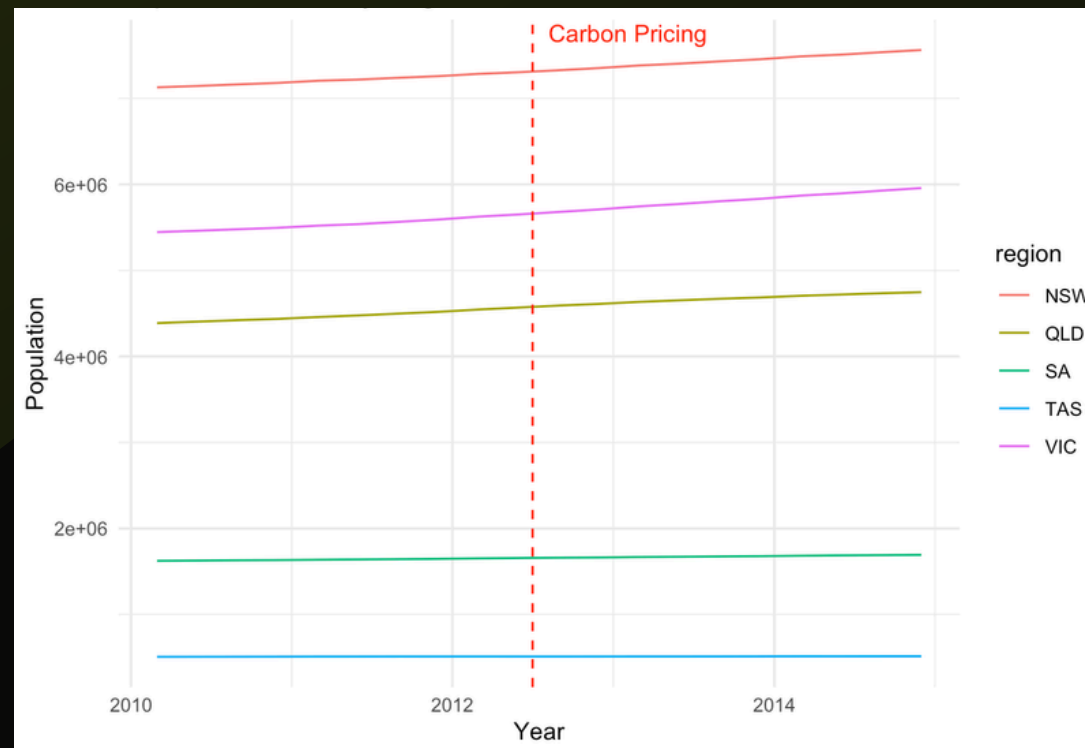
$temp_{it}$, $temp_{it}^2$, and $\log(pop_{it})$ are demand-side controls for seasonal temperature and state size.

Price excluded as it likely acts a mediator between carbon policy and demand, and due to endogeneity concerns

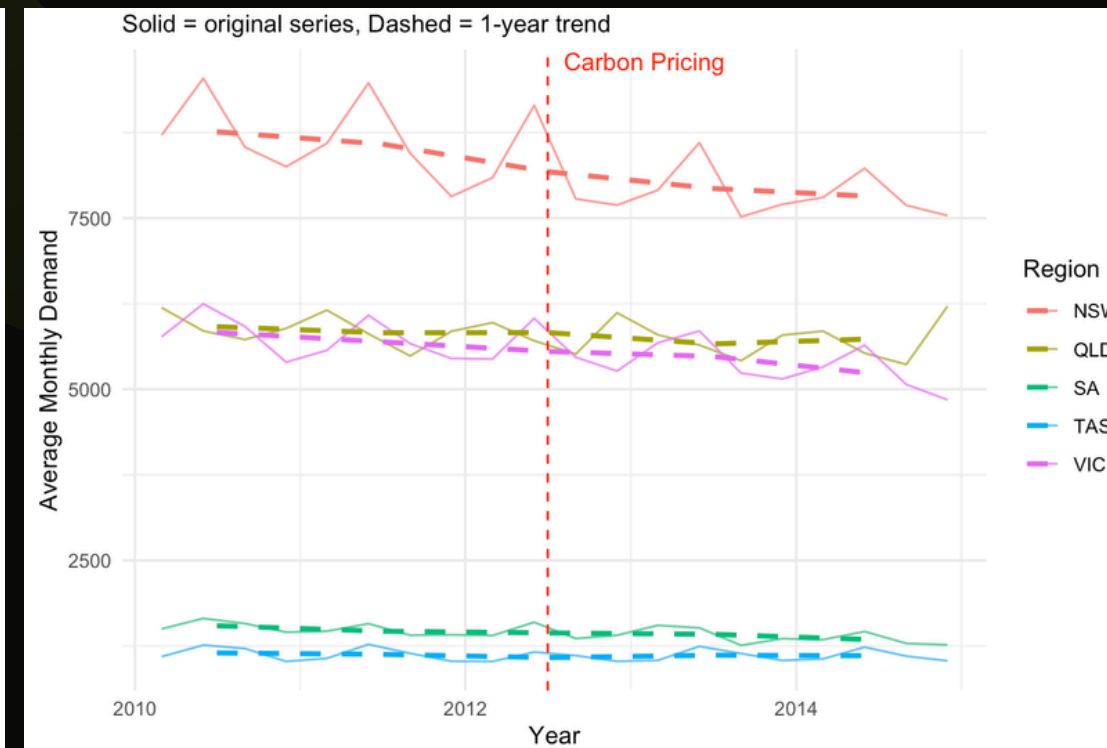
Coefficient are conditional correlation, not treatment effect



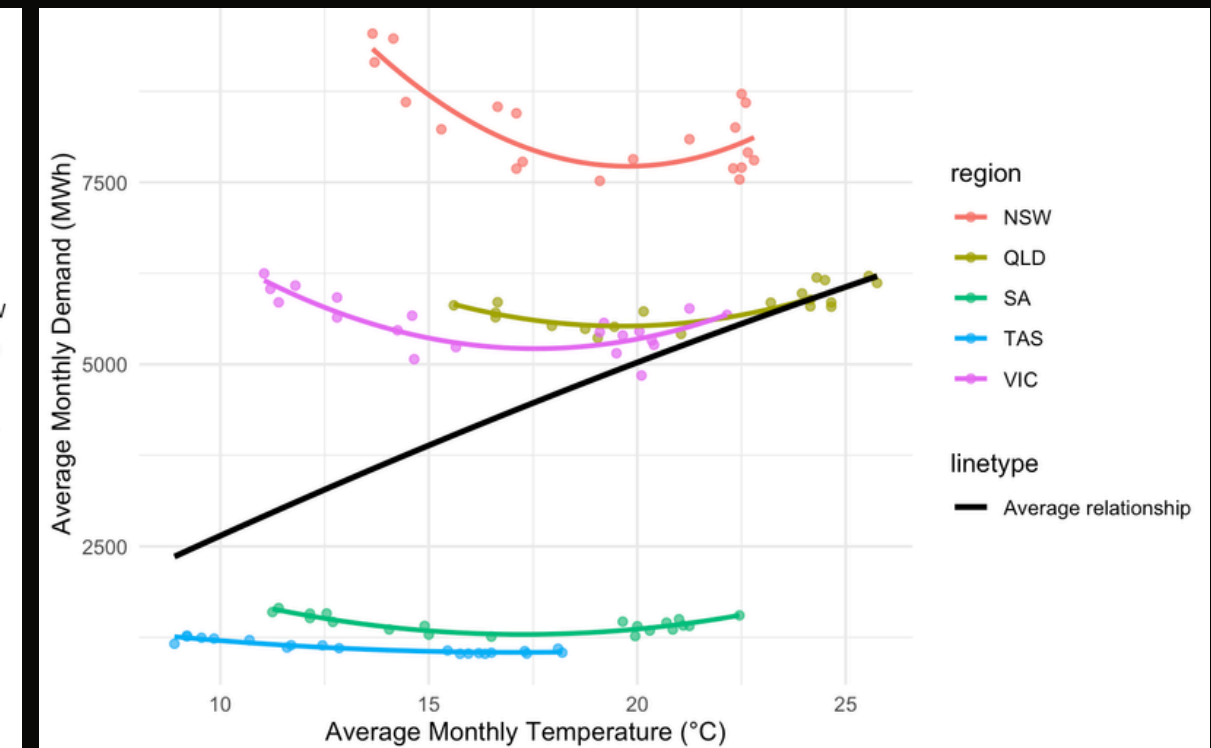
Specification Assumptions and Error Structure are Made Based on The EDA result



Population Trend



Demand Trend



Temperature and Demand Relationship

Wide ranges makes the log more economically meaningful

U-shape relationship motivated the quadratic form

Standard errors are clustered at the state level, as the assumption of independent errors is potentially violated in consecutive quarters.



The Evidence

3 specifications: **State FE + Time FE is preferred**

	Pooled	Within-State	Within-State & Time
	OLS	State FE	State+Time FE
Post x Coal Share	0.145 (0.138)	-0.016 (0.043)	-0.060 (0.054)
Temperature	-0.139*** (0.037)	-0.081*** (0.008)	-0.059** (0.014)
Temperature ²	0.004*** (0.001)	0.002*** (0.000)	0.002*** (0.000)
Num.Obs.	100	100	100
R ²	0.917	0.998	0.999
R ² Adj.	0.912	0.998	0.999

Dependent variable: log(avg quarterly electricity demand). Standard errors in parentheses: heteroskedasticity-robust for OLS; clustered by state for specifications (2) and (3). Intercept and post_jul2012 indicator omitted from display. The high R² is largely mechanical; state fixed effects absorb substantial cross-state variation in demand levels.

*** p<0.01, ** p<0.05, * p<0.1.

- Moving from a coal share of 0 to a coal share of 0.1 (e.g., from 0.7 to 0.8) is associated with approximately 0.6% lower electricity demand in the post-policy period relative to the pre-policy period, holding state fixed effects, time effects, temperature, and population constant.
- Coefficient moved from 0.145 to -0.016, to -0.060.
- The pooled positive result reflects a cross-state size confound
- The coefficient flips negative after state FE was added
- Key interest is **insignificant** (p > 0.1) in all specification
- Quadratic temperature is significant (p<0.01), consistent with dual heating and cooling load

Direction Stable, Significance Fragile

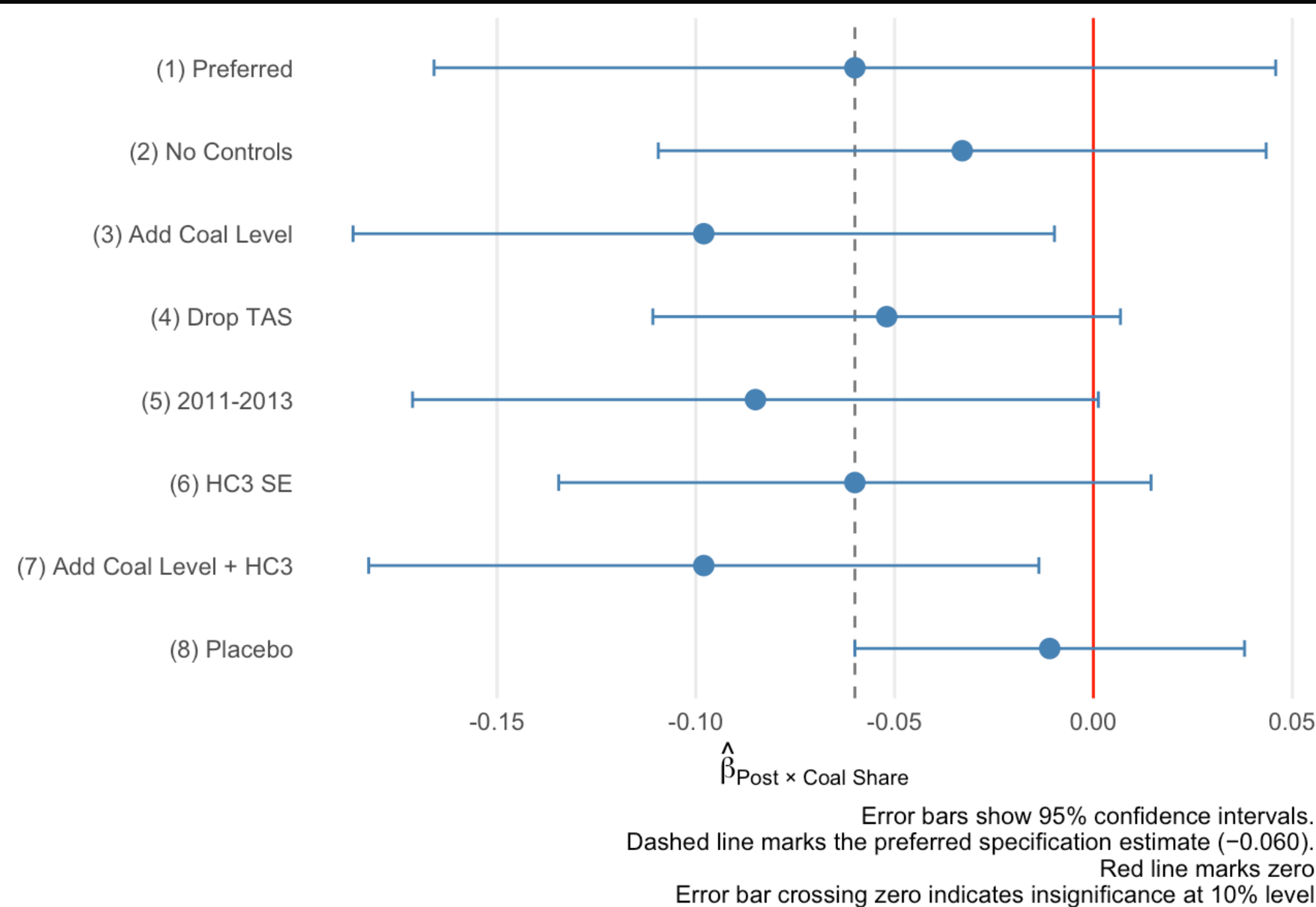
	Robust Check	Observation	Coefficient	Standard Error
Check whether the main result:	Preferred	100	-0.060	(0.054)
is sensitive to the set of controls?	No Control	100	-0.033	(0.039)
	Add Coal Share	100	-0.098*	(0.045)
is driven by a specific state or time window?	Drop Tasmania	80	-0.052	(0.030)
	Only 2011-2013	60	-.0085	(0.044)
is driven by the log transformation?	Demand Level Form	100	-852.119	(461.668)
is sensitive to the choice of standard error?	HC3	100	-0.060	(0.038)
	HCE + Add Coal Share	100	-0.098**	(0.043)
is driven by pre-existing trend rather than specific to the post-july period? Using 2010-2012 data	Placebo cut off at 2011	60	-0.011	(0.035)

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

The surviving and failing patterns suggest the main finding is credible in direction and economically meaningful in magnitude, but statistically sensitive to sample size and specification choices



Direction Stable, Significance Fragile



Direction stable

All 8 checks negative. Sign never flips. Consistent with genuine association.

Significance fragile

Significant only when coal share added as level term (10%) and when it combined with HC3 SEs (5%).

Placebo passes

Placebo = -0.011 (near zero). Association specific to post-July 2012 period.

Levels spec fails

Confirms log transformation is appropriate. Levels unstable due to scale differences.



The Limitations



4 Threats and Limitations

Time Varying Omitted Confounders

Several states introduced energy efficiency initiatives around the same period, and there is deindustrialisation. If these were more prevalent in high-coal states, this would bias the association downward.

Interstate Electricity Trade

High-coal states may have increased net imports after the carbon price made their generation expensive. Observed demand reduction could partly reflect generation substitution, not true consumption changes.

Measurement Constraints

Coal shares slow within-state variation, limiting the precision of the interaction estimate; quarterly aggregation smooths short-run demand responses.

Small sample

$N = 100$, $G = 5$ clusters. Cluster-robust SEs unreliable with only 5 clusters. Significance results should be interpreted cautiously. Inherent data constraint, not modelling failure.

Conclusion

- Small negative association: -0.060 (preferred), -0.098 (add coal level, significant at 5%)
- Direction is stable across all 8 robustness checks — never flips sign
- Statistically fragile: insignificance partly reflects near-collinearity, not absence of association
- Temperature is the standout result — U-shaped, significant at 1%, confirms quadratic specification
- Placebo test passes — association is specific to post-July 2012 period
- Finding consistent with known short-run inelasticity — energy demand adjusts slowly
- Suggestive, not conclusive. Longer time series, higher-frequency data, or a quasi-experimental design would resolve the remaining threats.



Thank You
for Your Attention